

Project Documentation

1. Introduction

Project Title: AI FAQ ASSISTANT API

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2. Project Overview

Purpose: Build a robust, secure, and intelligent REST API platform to streamline FAQ creation, user authentication, and AI-powered content management for customer support operations.

Key Features:

- JWT Authentication & Authorization for secure private endpoints
- AI FAQ Generation using Google Gemini AI (questions, answers, topics)
- Semantic / Keyword-based FAQ Search for fast information retrieval
- Complete CRUD operations for FAQ records
- Category management for structured organization
- Password encryption with bcrypt
- Centralized error handling and Mongoose schema validation
- RESTful API architecture with MongoDB + Mongoose ODM

3. Architecture

Backend Stack: Node.js + Express.js

Database: MongoDB with Mongoose ODM (Users, FAQs)

AI Integration: Google Gemini SDK for automated FAQ generation

Security: JWT tokens + bcrypt password hashing + auth middleware

Layered Architecture (MVC):

- **Model Layer** — Mongoose schemas with field validation and password hashing hooks
- **Controller Layer** — Request handling, business logic, delegation to services/models
- **View / Routing Layer** — Express routes mapping HTTP methods to controllers (headless API)
- **Middleware** — Auth token validation, centralized error handling, CORS

4. Software & Hardware Requirements

Software:

- OS: Windows 10/11, macOS, or Linux
- Node.js v18+, npm v9+, Express.js, MongoDB / Mongoose, Postman or Thunder Client
- Packages: express, mongoose, dotenv, cors, jsonwebtoken, bcrypt, @google/genai

Hardware: Intel Core i5 (8th Gen+) / AMD Ryzen 5, 8 GB RAM (16 GB recommended), 1 GB free disk space

5. Database Entities (MongoDB Collections)

User Entity

- `_id` (ObjectId), `name` (String, Required), `email` (String, Required, Unique)

- password (String, Hashed, Required), role (String, Default: "user"), createdAt, updatedAt

FAQ Entity

- _id (ObjectId), question (String, Required), answer (String, Required)
- category (String, Required), createdBy (FK → User), createdAt, updatedAt

AI Generation (Logical)

- topic, generatedQuestion, generatedAnswer, generatedCategory, generatedAt (produced by Gemini before persistence)

Category (Logical)

- categoryName, description, totalFAQs
- Key Relationships:**
- User → FAQ (One-to-Many): A user can create and manage multiple FAQ records
 - Category → FAQ (One-to-Many): Each category holds multiple FAQ documents
 - User → AI Generated FAQ (One-to-Many): Users can generate multiple AI-assisted FAQs

6. Roles and Responsibilities

Admin: Manage all users, view/update/delete any FAQ, monitor AI content, manage categories, full system access

Content Creator: Create FAQs, edit/delete own FAQs, generate FAQs with AI, organize under categories

Authenticated User: View public FAQs, search FAQs, generate AI-assisted answers, access personal profile

Public User: View published FAQs, search FAQs; cannot create/edit/delete; no access to protected endpoints

7. Setup Instructions

Prerequisites: Node.js v18+, MongoDB, npm, Git, Postman/Thunder Client

Installation Steps:

1. Create project folder and open in VS Code
2. Run: `npm init -y`
3. Install dependencies: `npm install express mongoose dotenv cors jsonwebtoken bcrypt @google/genai`
4. Configure environment variables (.env) for MongoDB URI, JWT secret, Gemini API key
5. Start server: `npm run dev`

8. Folder Structure

```
AI-FAQ-Assistant-API/
├── src/
│   ├── server.js           # Main entry - env, DB, port binding
│   ├── app.js              # Express app setup, CORS, routes
│   └── middleware/
│       ├── authMiddleware.js # JWT extraction & validation
│       ├── errorMiddleware.js # Centralized error handling
│       └── models/
│           ├── User.js      # User schema + password hash hook
│           └── FAQ.js       # FAQ schema with createdBy ref
│       └── routes/
│           ├── authRoutes.js # Register / Login
│           ├── faqRoutes.js  # Protected FAQ CRUD + search
│           └── aiRoutes.js   # AI generate-faq endpoint
│       └── controllers/     # Business logic handlers
```

■■■■ .env ■■■■

package.json

■■■■ README.md

9. API Documentation

Authentication (Public)

- **POST /api/auth/register** — Register new user (name, email, password)
- **POST /api/auth/login** — Login and receive JWT token

FAQ Operations (Protected – Bearer Token)

- **POST /api/faqs** — Create FAQ (question, answer, category)
- **GET /api/faqs** — List FAQs / public published articles
- **GET /api/faqs/search?q=keyword** — Semantic / keyword search (Public)
- **PUT /api/faqs/:id** — Update own FAQ
- **DELETE /api/faqs/:id** — Delete own FAQ

AI Automation (Protected)

- **POST /api/ai/generate-faq** — Generate FAQ pair from topic using Google Gemini

Payload example: { "topic": "Mongoose schema indexing validation" } **Sample**

Request Bodies:

Register:

```
{ "name": "Priya Sharma", "email": "priya@writeflow.com", "password": "securepassword123" }
```

Create FAQ:

```
{ "question": "How do I configure custom category tags?", "answer": "Navigate to settings...",  
  "category": "Configuration" }
```

10. Authentication

JWT-based login system. On successful login the server returns a signed token. Protected routes require the header `Authorization: Bearer <token>`. Auth middleware extracts the token, verifies the signature, and attaches user context to the request. Passwords are hashed with bcrypt before storage. Only original authors (or admins) can modify their documents.

11. Running the Application

- Ensure MongoDB is running and .env is configured
- Start the development server: `npm run dev`
- Test endpoints using Postman or Thunder Client
- Typical base URL: `http://localhost:5000` (or configured PORT)

12. Scenario & User Flow

Background: Priya, a customer support operations manager, struggles with repetitive inquiries, organizing entries, classifying topics, and tracking question trends.

Solution: The AI FAQ Assistant API provides a centralized, secure backend that decouples business logic from frontend systems. Authenticated creators register, generate AI-assisted FAQs, draft official schemas, and manage their records. Public users can browse and search published FAQs.

Outcome: Operational delays are minimized. Schema validation maintains data integrity, JWT tokens ensure only original authors modify documents, and indexing enables real-time discoverability.

13. Testing

- **Unit Testing** — Verified individual modules (Auth, FAQ CRUD, AI generation, Search)
- **Integration Testing** — Checked smooth interaction between routes, middleware, models and MongoDB
- **API Testing** — Validated all endpoints (register, login, create/search FAQ, AI generate) via Postman

• **Security Testing** — Confirmed protected routes reject unauthenticated requests; password hashing works • **Result**

— Application runs successfully with correct data flow and secure access control

14. Demo & Resources

- **API Testing Video:** https://drive.google.com/file/d/1u_w1sigiOynhmqTDbj4NBm3CLIMunEWN/view?usp=sharing
- **Demo Link:** https://drive.google.com/file/d/1ajAZGgOaowPdyW2x5otcsLI84NeomSYJ/view?usp=drive_link
- **Code Drive:** https://drive.google.com/drive/folders/1aUqV5q25fE_iUPwWcLt02eJ-v-ytxpjz?usp=sharing

15. Known Issues

- Currently a pure backend REST API — no built-in frontend UI
- AI generation depends on external Google Gemini API availability and quota
- Semantic search is primarily keyword-based; advanced vector search not yet implemented
- Role-based access control is present but advanced admin dashboard features are limited
- No multi-tenant / multi-organization isolation in the current schema

16. Future Enhancements

- Advanced vector / embedding-based semantic search for better relevance
- Frontend dashboard (React) for non-technical content creators
- Multi-tenant support and organization-level FAQ isolation
- Analytics on frequently asked questions and search trends
- Rate limiting, API key management, and usage quotas
- Webhook / notification support when new FAQs are published
- Export FAQs to Markdown / PDF / knowledge-base formats
- Fine-tuned domain-specific AI models for higher quality generation